

Public Economics Practicum 5

Discussion Questions Rosen:

	8th edition	9th & 10th editions
Chapter 14	3, 5, 6, 8	3, 5, 6, 8
Chapter 15	5, 7, 8	9, 11, 12

Rosen 14.3 will be discussed in the tutorial.

Additional Exercises:

5.1 The demand for duck-tape is given by $P = 10 - \frac{2}{3}Q$, the supply by $P = 1 + \frac{1}{3}Q$.

- (a) Find equilibrium price, quantity, and elasticities of demand and supply in equilibrium.

An unit tax t is imposed on suppliers. Suppose $t = 3$.

- (b) Find the new equilibrium. What is the incidence of the tax? How is this related to the elasticities of demand and supply you found under (a)? Why is tax incidence related to elasticities?
- (c) Find the excess burden of the tax. How do you think the excess burden depends on elasticities of demand and supply?

5.2 Betty (see exercise 4.3) has moved to another country. In this country, she can earn a hourly wage of 20 euro. The total number of hours available is 720, and she has to decide how to divide these between work and leisure. Of course, her preferences have not changed: Betty's utility of net income (in euro's) Y and leisure (in hours) L equals

$$U = Y^{\frac{1}{6}} L^{\frac{5}{6}}$$

- (a) How many hours does Betty work?

Now suppose that the government introduces a proportional income tax: $T = 0.2I$, where I is gross income, so that $Y = I - T$.

- (b) How many hours does Betty work? What is the tax revenue?

- (c) Determine the income and the substitution effect of the income tax on the consumption of leisure. (Hint: compute utility under (b). Then find out which combination of L and Y would yield this level of utility if relative prices of Y and L would not have changed, by computing the level of a lump-sum tax that yields the utility under b).
- (d) What is the excess burden of the tax? (Note: You need to compute the equivalent variation. Figure 15.3 may help).

5.3 Harry has a gross hourly wage of 20 euro and can divide 500 hours each month between leisure and work. He has diminishing marginal utility of both leisure and income. Without redistribution, Harry would work 200 hours per month. Now suppose that the country where Harry works, has the following system of redistribution: Everyone gets 600 euro from the government. This is paid by a 15% tax on gross income from work.

Argue whether Harry works more, less, or the same number of hours due to the system of redistribution, as compared to the 200 hours he would work without redistribution.

5.4 Suppose there are 100 equivalent individuals in a village. The mayor of the village is considering building a park in the centre of the village. Construction of the park costs 800. The utility each individual derives from the park will be equivalent to 10. To pay for the construction of the park, the mayor would introduce a unit tax t on gasoline. Individuals' demand for gasoline can be given by $q_i = 16 - 2p$. Supply of gasoline is perfectly elastic: $p = 4$.

- (a) Suppose construction cost could be paid for by a lump-sum tax. Is it efficient to construct the park?
- (b) Show that to raise 800, the tax t needs to be 2.
- (c) What would be the excess burden of this tax?
- (d) Should the park be constructed?

Now suppose that it is possible to vary the size of the park, which also affects the cost. Suppose the utility each individual derives from the park equals $\frac{1}{80}C$, where C is the total amount of money invested in the park.

- (e) Determine the optimal size of investment in the park C .

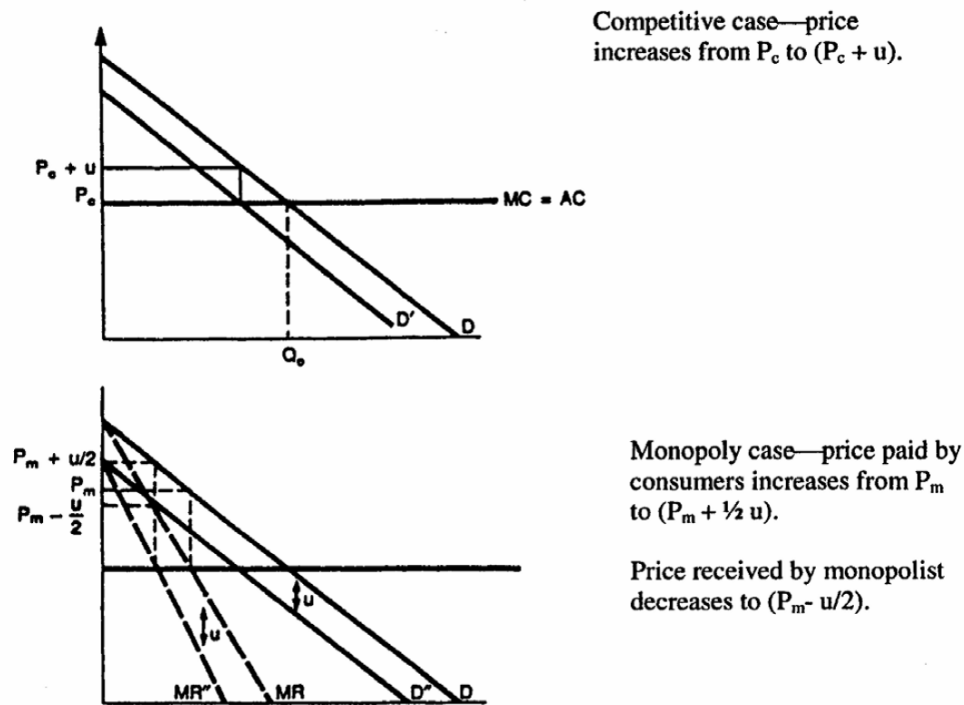


Figure 1: Rosen 14.3

Answers to exercises week 5

Rosen 14.3: See figure

Rosen 14.5

- Before-tax equilibrium: $P = \$10$ and $Q = 300,000$. After-tax equilibrium: $P = \$10.60$ and $Q = 288,000$. Consumers pay $\$10.60$ and producers receive $\$9.60$.
- Revenue = $\$288,000$. Consumers bear 60 percent of the tax burden and producers bear 40 percent. So, $\$172,800$ comes from consumers and $\$115,200$ from producers.
- With a more elastic demand curve, quantity consumed will decrease even more as a result of the tax, so the liquor tax will be more effective at reducing consumption among young drinkers.

Rosen 14.6

You can view this exercise as proving (for the case of linear demand and supply curves) that tax incidence does not depend on whether the tax is levied on consumers

or producers.

Without tax, the equilibrium price can be calculated by equating demand and supply

$$a - bP = c + dP \rightarrow P = \frac{a - c}{b + d}$$

The equilibrium output can be determined by substituting the equilibrium price into either the supply or demand equation, which gives

$$Q = \frac{ad + bc}{b + d}$$

Now suppose that a unit tax of u dollars is imposed. We will show that it doesn't matter which party it is imposed upon (the consumer or producer); the new equilibrium will be the same in either case. First, suppose the unit tax is imposed upon the consumer. Then, the effective demand curve changes into $Q^{DT} = a - b(P + u)$. Note, the price P is here defined as the price consumers pay to producers. On top of this price, consumers pay tax u . Equating demand Q^{DT} and supply Q^S gives

$$a - b(P + u) = c + dP \rightarrow P = \frac{a - c - bu}{b + d}$$

This is the price producers receive. Consumers pay $P + u$, or

$$\frac{a - c - bu}{b + d} + u = \frac{a - c + du}{b + d}$$

Equilibrium quantity follows from substituting P into Q^{DT} or Q^S

$$Q^{DT} = a - b \left(\frac{a - c + du}{b + d} \right) = \frac{cb + da - dbu}{b + d}$$

Note: Compared to the case without tax, producers receive a lower price, consumers pay a higher price, and quantity is reduced.

Now suppose that the tax is imposed on producers. Then, the effective supply curve changes to $Q^{ST} = c + d(P - u)$. Again, note that the price P is here defined as the price consumers pay to producers. Producers have to pass on u to the government. Equating demand Q^D and supply Q^{ST} gives

$$a - bP = c + d(P - u) \rightarrow P = \frac{a - c + du}{b + d}$$

This is what consumers pay to producers. The producers keep $P - u$, or

$$\frac{a - c + du}{b + d} - u = \frac{a - c - bu}{b + d}$$

Equilibrium quantity is given by

$$Q^{ST} = c + d\left(\frac{a - c - bu}{b + d}\right) = \frac{bc + ad - dbu}{b + d}$$

Hence, we find the same prices and quantity.

Rosen 14.8

Progressivity depends on the average tax rate. In the tax scheme $T = -4000 + 0.2I$, the percentage of income I paid in taxes is $ATR = -\frac{4000}{I} + 0.2$. The best way to determine whether the tax scheme is progressive or regressive is to see what happens to ATR if I increases. Taking the derivative of ATR with respect to I gives

$$\frac{dATR}{dI} = \frac{4000}{I^2} > 0$$

Hence, the tax scheme is progressive.

A general scheme of this sort is $T = a + tI$. Here, average tax rate is $ATR = \frac{a}{I} + t$. The effect of I on ATR equals

$$\frac{dATR}{dI} = -\frac{a}{I^2}$$

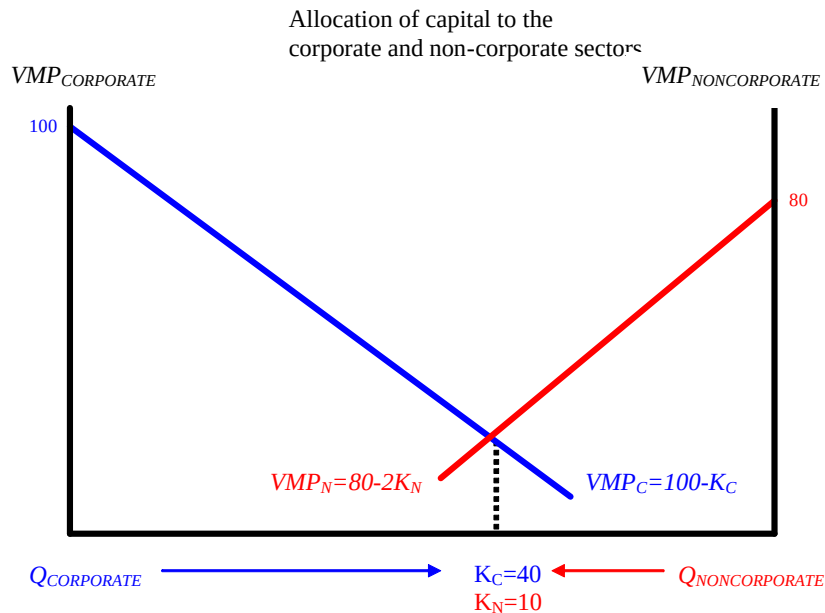
Hence, this tax scheme is progressive if $a < 0$ and regressive if $a > 0$.

Rosen 15.5 / 15.9

It is likely that the elasticity of demand for television is quite inelastic. It follows that the excess burden from a \$160 per year television tax is small relative to the revenues that are collected.

Rosen 15.7 / 15.11

a) The value of the marginal product of capital in the corporate sector is given by $VMP_c = 100 - K_c$, and the value of the marginal product of capital in the noncorporate sector is given by $VMP_n = 80 - 2K_n$. Without taxation, capital will be allocated such that the marginal product is equal in both sectors (otherwise an owner of a unit of capital in the sector where the marginal product is lowest can gain by moving the unit to the other sector). Thus, setting $VMP_c = VMP_n$ gives $100 - K_c = 80 - 2K_n$. Substituting for $K_c + K_n = 50$ gives $100 - 50 + K_n = 80 - 2K_n$, implying that $K_n = 10$ and $K_c = 40$.

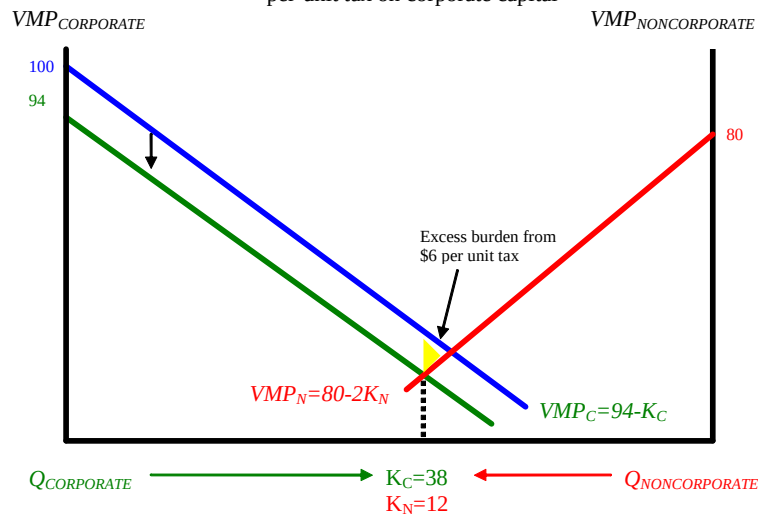


Rosen 15.7a

b) If a unit tax of \$6 is leveled on capital employed in the corporate sector, the after-tax value of the marginal product in the corporate sector falls. It is now given by $VMP'_c = 100 - K_c - 6 = 94 - K_c$. Setting $VMP'_c = VMP_n$ gives $94 - K_c = 80 - 2K_n$. Substituting for $K_c + K_n = 50$ gives $94 - 50 + K_n = 80 - 2K_n$, implying that $K_n = 12$ and $K_c = 38$.

As can be seen in the figure, the excess burden of this tax is a triangle with height 6 (the tax) and base 2 (the amount of capital shifted from the corporate to the noncorporate sector due to the tax). Hence, the tax burden is $\frac{1}{2} \cdot 6 \cdot 2 = 6$.

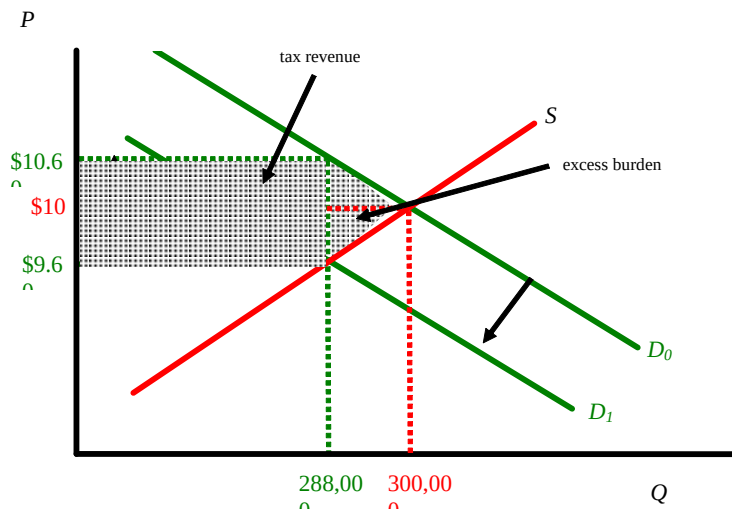
FIGURE 13.7b – Reallocation after per-unit tax on corporate capital



Rosen 15.7b

Rosen 15.8 / 15.12 (relates to 14.5)

a) Before-tax equilibrium: $P = \$10$ and $Q = 300,000$. After-tax equilibrium: $P = \$10.60$ and $Q = 288,000$. Consumers pay \$10.60 and producers receive \$9.60. Excess Burden = $\frac{1}{2}(12,000)(\$0.60) + \frac{1}{2}(12,000)(\$0.40) = \$6,000$.



Rosen 15.8

If the negative external cost were equal to \$1 per gallon, then a \$1 tax would achieve an efficient allocation and would create no excess burden. With a negative external cost of \$0.50 per gallon, there is still an excess burden associated with a \$1 per gallon tax, but it is smaller since the efficient level of output in this market would be between 288,000 and 300,000.

Exercise 5.1

a) Equilibrium $Q = 9$, $P = 4$. Demand elasticity:

$$\varepsilon^d = \frac{dQ^d}{dP} \frac{P}{Q^d}$$

To find $\frac{dQ^d}{dP}$, either rewrite demand function to $Q^d = 15 - 1.5P$ and take the derivative to P , or simply take the derivative of $P = 10 - \frac{2}{3}Q^d$ to Q^d (Note: this gives $\frac{dP}{dQ^d}$) and use $\frac{dQ^d}{dP} = \frac{1}{dP/dQ^d}$. Both give $\frac{dQ^d}{dP} = -1.5$. Hence, in equilibrium, $\varepsilon^d = \left| -\frac{2}{3} \right| = \frac{2}{3}$. Similarly, $\varepsilon^s = \frac{dQ^s}{dP} \frac{P}{Q^s} = \frac{4}{3}$.

b) Effective supply function is now $P - 3 = 1 + \frac{1}{3}Q^s$ or $P = 4 + \frac{1}{3}Q^s$. New equilibrium: $Q = 6$, $P = 6$. So consumers pay 6, producers keep 3. Tax revenue equals $3 \cdot 6 = 18$. For consumers, price has increased by 2, for producers, the price has decreased by 1. Hence, consumers effectively pay 12 in taxes, producers 6. Consumers thus bear two-thirds of the tax burden, producers one-third. It is no coincidence that the elasticity of supply is twice as large as the elasticity of demand in the pretax equilibrium and that consumers pay twice as much in taxes as producers. In general, the distribution of the tax burden over consumers and producers depends on the relative elasticities of demand and supply. Equal elasticities (ignoring the difference in sign) imply an equal split, and the more elastic one curve is compared to the other, the smaller the share of the tax burden this group has to bear. This reflects that relative quality of alternatives for consumption and production: if consumers can easily switch to consuming another good, but producers have few alternative production options, producers will end up bearing most of the tax burden.

c) Excess burden is $\frac{1}{2}t\Delta Q = \frac{1}{2}3(9 - 6) = 4\frac{1}{2}$. Note that this implies that to raise 18 euro, consumers lose 15 and producers lose 7.5. As discussed in week 6: if elasticities are larger, the reduction in equilibrium quantity from a given tax is larger, which implies that the excess burden is larger.

Exercise 5.2

a) Maximal earning are $20 \cdot 720 = 14400$. This gives budget constraint $Y = 14400 - 20L$. Hence, Lagrange equation is

$$\Gamma = Y^{\frac{1}{6}} L^{\frac{5}{6}} + \lambda (14400 - 20L - Y)$$

Differentiating to Y , L , and λ gives

$$\begin{aligned}\frac{\partial \Gamma}{\partial Y} &= \frac{1}{6}Y^{-\frac{5}{6}}L^{\frac{5}{6}} - \lambda = 0 \\ \frac{\partial \Gamma}{\partial L} &= \frac{5}{6}Y^{\frac{1}{6}}L^{-\frac{1}{6}} - 20\lambda = 0 \\ \frac{\partial \Gamma}{\partial \lambda} &= 14400 - 20L - Y = 0\end{aligned}$$

Combining the first two gives $\frac{1}{6}Y^{-\frac{5}{6}}L^{\frac{5}{6}} = \frac{5}{120}Y^{\frac{1}{6}}L^{-\frac{1}{6}}$, which can be rewritten to $L = \frac{1}{4}Y$. Substituting this into the third derivative gives $14400 - 6Y = 0$. Hence, $Y = 2400$ and $L = 600$.

b) Now, maximal net income is $(1 - 0.2) \cdot 20 \cdot 720 = 11520$. Budget constraint is now $Y = 11520 - 16L$. Lagrange equation is

$$\Gamma = Y^{\frac{1}{6}}L^{\frac{5}{6}} + \lambda(11520 - 16L - Y)$$

Differentiating to Y , L , and λ gives

$$\begin{aligned}\frac{\partial \Gamma}{\partial Y} &= \frac{1}{6}Y^{-\frac{5}{6}}L^{\frac{5}{6}} - \lambda = 0 \\ \frac{\partial \Gamma}{\partial L} &= \frac{5}{6}Y^{\frac{1}{6}}L^{-\frac{1}{6}} - 16\lambda = 0 \\ \frac{\partial \Gamma}{\partial \lambda} &= 11520 - 16L - Y = 0\end{aligned}$$

Combing the first two gives $\frac{1}{6}Y^{-\frac{5}{6}}L^{\frac{5}{6}} = \frac{5}{96}Y^{\frac{1}{6}}L^{-\frac{1}{6}}$. This can be rewritten to $L = \frac{30}{96}Y$. Substituting into the third derivation gives $11520 - 6Y = 0$. Hence, $Y = 1920$ and $L = 600$. Tax revenue is $0.2 \cdot 20(720 - 600) = 480$.

c) With $Y = 1920$ and $L = 600$, her utility equals

$$U = (1920)^{\frac{1}{6}}(600)^{\frac{5}{6}} \approx 728.35$$

In case of a lump-sum tax T , Betty's budget constraint is $Y = 14400 - T - 20L$. However, relative prices of income and leisure are identical to those under (a). Hence, only the last derivative of the Lagrange under (a) is affected by this lump-sum tax. This implies that from the first two, we have again that $L = \frac{1}{4}Y$ or $4L = Y$. Substituting that into the utility function, we get

$$U = (4L)^{\frac{1}{6}}(L)^{\frac{5}{6}} = 4^{\frac{1}{6}}L$$

If we set this equal to 728.35, we are on the same indifference curve as under (b). $4^{\frac{1}{6}}L = 728.35$ gives $L \approx 578.1$. Hence, $Y = 4 \cdot 578.1 \approx 2312.37$. Hence, the income

effect of the tax is to reduce leisure by about 21.9 hours. The substitution effect of the tax (which makes leisure relatively less expensive) increases leisure by the same amount.

d) Even though Betty keeps working the same hours, there is an excess burden, as the excess burden stems from the substitution effect. The easiest way to see that there is a excess burden is by imposing a lump-sum tax of 480 euro on Betty (rather than the 20% income tax). Then, budget constraint is $Y = 14400 - 480 - 20L$. Hence, compared to (a), the only difference is that 14400 is changed into 13920. Hence, with a lump-sum tax of 480, Betty's optimal Y and L are $Y = 2320$ and $L = 580$. The fact the we find something different under (b) indicates that there is an excess burden to this tax, even though there is no change in L .

To find the excess burden of the income tax, we can use the outcomes derived under c). Substituting for $L \approx 578.1$ and $Y = 2312.37$ into the budget constraint with lump-sum tax T (recall: $Y = 14400 - T - 20L$), we find $T \approx 525.63$. This means that if we could levy a lump-sum tax rather than the 20% income tax, a lump-sum tax of 525.63 would make Betty as well off as the 20% income tax. However, the latter only raises 480 in revenue. Hence, the excess burden of the income tax is $525.63 - 480 \approx 45.63$. This is almost 10% of tax revenue.

Exercise 5.3

Without redistribution, Harry has 300 hours of leisure and 4000 euro of income. In that optimal point, Harry is indifferent between an additional hour of leisure and an additional hour of work (indifference-curve touches budget constraint). An hour of work yields 20 euro, so that the marginal value of an hour of leisure in that point is also 20 euro. With redistribution, the same point can be attained: If Harry works 200 hours, his income is $600 + 0.85(20 \cdot 200) = 4.000$ euro. Hence, he could choose exactly the same point. However, with redistribution, an hour of work only yields net 17 euro. From the above argumentation, his value of an hour of leisure, given that his income is 4000 euro and he has 300 hours of leisure, is 20 euro. Hence, Harry will optimally work less.

This is most easily seen when drawing the old and the new budget constraints: the new budget constraint rotates exactly at the optimal point on the old budget constraint. Hence, it must be that the indifference curve that exactly touches the old budget constraint at 300 hours of leisure lies under the new budget constraint for a range of values $L > 300$. Hence, the optimal new point lies on the part of the new budget constraint where $L > 300$.

Exercise 5.4

a) Yes. If it would be possible to levy a lump-sum tax, there is no excess burden. Hence, total cost of the park equals 800, whereas total benefit equals 1000.

b) Perfectly elastic supply at $p = 4$ implies that the effective price including unit tax t consumers pay equals $p^c = 4 + t$. At this price, individual demand equals $q_i = 16 - 2(4 + t)$. Total tax revenue should be 800. Two ways to approach. (1) individuals are identical and, hence, will pay the same amount in taxes. Hence, each individual has to pay 8 in taxes, or $tq_i = 8$. This gives

$$tq_i = 16t - 8t - 2t^2 = 8 \rightarrow 4t - t^2 - 4 = 0$$

which can be solved for $t = 2$.

(2) Total demand equals $100q_i$. Hence, tax revenue is $t \cdot 100q_i$, which must be equal to 800. This gives

$$t \cdot 100q_i = 1600t - 800t - 200t^2 = 800 \rightarrow 4t - t^2 - 4 = 0$$

which can be solved for $t = 2$.

c) Without tax, each individual buys $q_i = 8$. With $t = 2$, we get $q_i = 4$. Hence, for each individual, excess burden is $\frac{1}{2}t\Delta q_i = \frac{1}{2} \cdot 2 \cdot 4 = 4$. Total excess burden is $100 \cdot 4 = 400$.

d) No. Now, total economic cost of building the park is $800 + 400 = 1200$. Total benefits are only 1000. Hence, welfare decreases if the park is constructed. Note: you can also determine the total cost of the tax (tax revenue + EB) as the reduction in consumer surplus in the market for gasoline. Before tax, $CS = 100 \cdot \frac{1}{2} (8 - 4) 8 = 1600$. After tax, $CS = 100 \cdot \frac{1}{2} (8 - 6) 4 = 400$. Hence, building the park would yield 1000 in wellbeing related to the park, but costs 1200 in wellbeing related to consuming gasoline.

e) Using b), tax t yields total revenues equal to $t \cdot 100q_i = 1600t - 800t - 200t^2$. This is also total investment in the park. Excess burden equals $100 \cdot \frac{1}{2}t\Delta q_i = 100 \cdot \frac{1}{2}t(16 - 2(4) - (16 - 2(4 + t))) = 100t^2$. Hence, total net benefit of investing in the park equals Benefit - Cost - EB = $100 \cdot \frac{1}{80} (1600t - 800t - 200t^2) - (1600t - 800t - 200t^2) - 100t^2 = -50t(3t - 4)$. Maximising this wrt t yields first-order condition $200 - 300t = 0 \rightarrow t = \frac{2}{3}$. Hence, total investment in the park is optimally $800 \left(\frac{2}{3}\right) - 200 \left(\frac{2}{3}\right)^2 = \frac{4000}{9} = 444\frac{4}{9}$.